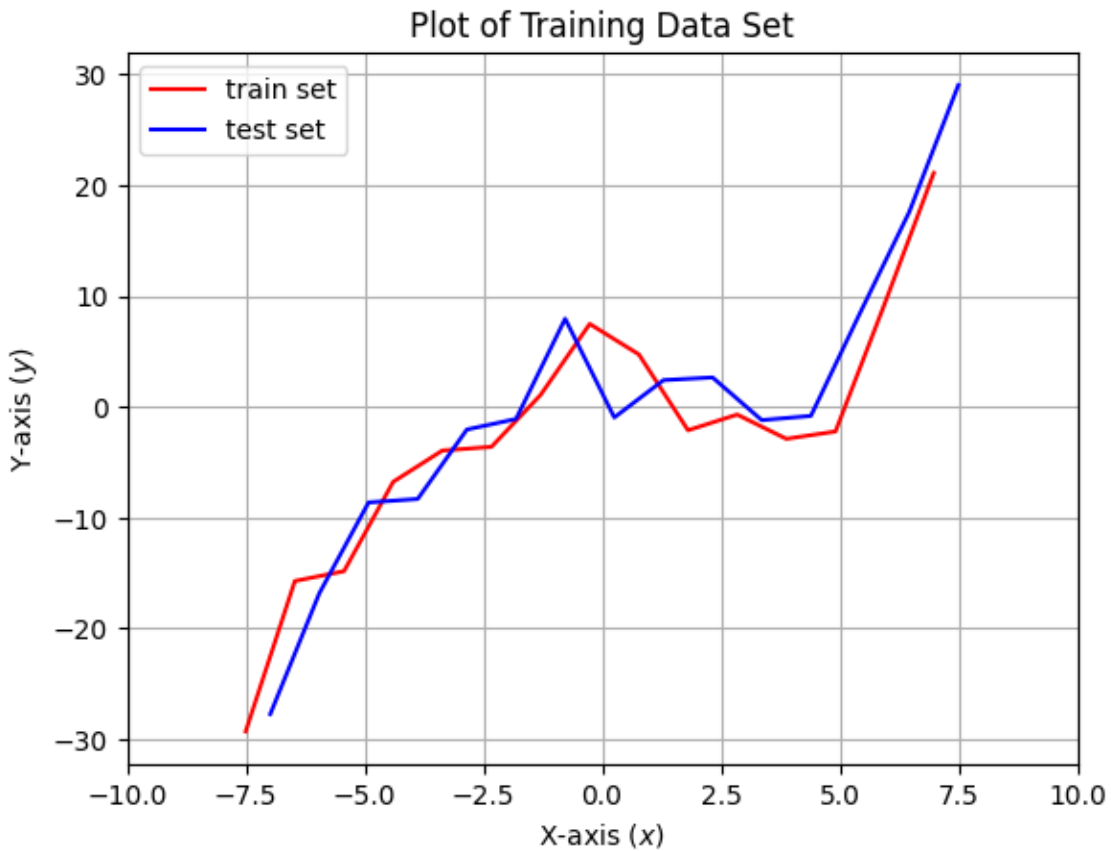


Step 1: Plotting the relationships between Training and Test sets

```
plt.figure()
plt.plot(x_train, y_train, color='r', label='train set')
plt.plot(x_test, y_test, color='b', label='test set' )
plt.xlabel("X-axis ($x$)")
plt.ylabel("Y-axis ($y$)")
plt.title("Plot of Training Data Set")
plt.xlim([-10, 10]) # Setting x-axis limits
plt.legend()
plt.grid(True)
plt.show()
```



Step 2: cos_poly_feature define

```
def cos_poly_features(X, K):  
    X = X.flatten()  
    N = X.shape[0]  
    Phi = np.zeros((N, K+2)) # K+2 columns: K+1 polynomial + 1 cosine  
    for k in range(K+1):  
        Phi[:,k] = X**k  
    Phi[:,K+1] = np.cos(0.7 * X) # Add cosine feature  
    return Phi
```